

# When gold disconnects

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## Abstract

In the hire graph of the 3-free door, every odd prime other than 3 meets 5 in the infinite undirected gold graph. Finite windows can still be disconnected: each new 2-power-door prime is gold-isolated at its first-owner window. Downward gold chains terminate only at 2-power doors, so gold is disconnected for infinitely many finite  $X$  if and only if there are infinitely many Mersenne or Fermat primes. The companion note settles the infinite component of 5; this note records the equivalence for the leftover finite-window question, together with thin computational certificates for the  $M_{31}$  corridor.

## 1 Setup

We use the notation of the hire-graph note [1]: the 3-free door  $m_0$ , the hire set  $S_X$ , directed gold arcs  $q \mid m_0(p)$ , and the undirected gold graph on  $S_X \setminus \{2\}$ . A prime  $M \neq 3$  is a *2-power door* if  $m_0(M) = 2^k$  for some  $k \geq 1$ . Every Mersenne prime  $M_p = 2^p - 1$  with  $p$  odd, and every Fermat prime greater than 3, is of this form. The first owner  $r_0$  of  $M$  is the least prime  $r$  with  $M \mid m_0(r)$ ; by the door bound,  $r_0 \geq 2M - 1$ .

## 2 First-owner windows

**Lemma 1.** *Let  $M > 5$  be a 2-power-door prime and let  $r_0$  be its first owner. At the window  $X = r_0$ , the prime  $M$  is an isolated vertex of undirected gold. In particular gold is disconnected.*

*Proof.* Since  $M \mid m_0(r_0)$  and  $r_0 \leq X$ , one has  $M \in S_X$ . The first owner itself is not hired at this window: nothing smaller than  $r_0$  can have hired it, so  $r_0 \notin S_X$ . As a 2-power door,  $M$  has gold out-degree zero. No other owner of  $M$  exists below  $r_0$ , so no incoming gold arc is live either. Thus  $M$  meets no other odd hired prime by a gold edge. Since  $M > 5$  and  $r_0 \geq 2M - 1 > 11$ , we also have  $5 \in S_{r_0}$  (for example  $5 \mid m_0(11) = 10$ ), so  $M$  is an isolated vertex distinct from another vertex of the gold graph.  $\square$

As of September 2026, every known Mersenne prime from  $M_{31}$  onward has first owner larger than the connecting window  $X^* = 92\,274\,421$  of the hire-graph note, so Lemma 1 already supplies many concrete disconnected windows. That is a finite list, not an infinitude theorem.

## 3 Downward chains

**Lemma 2** (Door closure in a window). *If  $q \in S_X \setminus \{2\}$  and  $r$  is an odd prime dividing  $m_0(q)$ , then  $r \in S_X$ .*

*Proof.* Every odd hired prime satisfies  $q \leq (X+1)/2 \leq X$ , so  $q$  itself is an owner in the window. Any odd prime factor of  $m_0(q)$  is therefore hired into  $S_X$ .  $\square$

**Lemma 3.** *Let  $S$  be a finite nonempty set of odd primes other than 3, closed under taking odd prime factors of  $m_0$ . Then  $S$  contains a 2-power door.*

*Proof.* Take a least element  $p \in S$ . If  $m_0(p) = 2^k$ , we are done. Otherwise some odd prime  $q \neq 3$  divides  $m_0(p)$ . Closure puts  $q \in S$ , and the size bound  $q \leq m_0(p)/2 < (p+1)/2$  forces  $q < p$ , contradicting minimality.  $\square$

Gold arcs on odd leaves are strictly decreasing in the prime label, so every finite descending gold chain ends at a sink, and Lemma 3 says every such sink in a door-closed set is a 2-power door.

## 4 The equivalence

**Theorem 1.** *The following are equivalent.*

1. *There are infinitely many finite windows  $X$  on which undirected gold is disconnected.*
2. *There are infinitely many 2-power-door primes.*
3. *There are infinitely many Mersenne primes, or infinitely many Fermat primes.*

*Proof.* Lemma 1 gives  $(2) \Rightarrow (1)$ : each sufficiently large 2-power door contributes at least its first-owner window. For  $(1) \Rightarrow (2)$ , suppose only finitely many 2-power doors exist. After the last of them has been absorbed into the infinite component of 5 (which the hire-graph note supplies for every odd prime other than 3), every later hired vertex has a downward gold path onto an already-absorbed sink. That path is live in the window by Lemma 2. Hence later windows stay connected. The equivalence of (2) and (3) is the classification of 2-power doors as Mersenne or Fermat primes greater than 3.  $\square$

Fermat primes are widely believed finite (five known; the last is 65537). Infinitude of Mersenne primes is open. Thus Theorem 1 does not settle item (1) unconditionally; it identifies the obstruction.

**Remark 1** (Heuristic). *The Lenstra–Pomerance–Wagstaff heuristic predicts about  $e^\gamma \log_2 x$  Mersenne primes with exponent at most  $x$ , hence infinitely many, and therefore infinitely many disconnected windows. (An additional, separate heuristic is that each large Mersenne prime opens a satellite interval whose length is governed by owner and hire delays in residue classes modulo a multiple of  $M_p$ ; that is not a consequence of LPW alone.) That is a labelled heuristic, not a proof of (1).*

## 5 The $M_{31}$ integers

Let  $M_{31} = 2^{31} - 1$ . Thin CRT searches (exhaustive over the two admissible classes, not a gold-graph rebuild) yield the following computational certificates:

$$r_0 = 98\,784\,247\,763, \quad v = 300\,647\,710\,579,$$

where  $r_0$  is certified as the least prime owner of  $M_{31}$  in either class modulo  $3M_{31}$ , and  $v$  is certified as the least prime in either class modulo  $15M_{31}$  with  $5M_{31} \mid m_0(v)$  (class A;  $m_0(v) = 2^2 \cdot 5 \cdot 7 \cdot M_{31}$ ). The ratio  $v/r_0 \approx 3.04$ . The floor candidate  $2M_{31} - 1$  is composite ( $9241 \cdot 464773$ ); the certificate for  $r_0$  is that no earlier candidate in those two classes is prime.

At  $X = v$  the integer  $v$  is an owner, so 5 and  $M_{31}$  divide its door, but  $v \notin S_X$ , and the gold arcs  $v \rightarrow 5$  and  $v \rightarrow M_{31}$  are not live until  $v$  itself is hired. That hiring time is at least  $2v - 1 \approx 6.01 \cdot 10^{11}$ . What  $v$  records is a bridge owner, not yet a bridge vertex. A mixed owner hired earlier than  $v$  could still connect  $M_{31}$  sooner; “first direct  $5$ – $M_{31}$  bridge” is not automatically the first connecting vertex. A full component check at these scales remains past home computing.

## Lean

Lemmas 1–3 and Theorem 1 are checked in Lean 4 (`Hire/GoldDisconnects.lean`; declaration names on the companion site). Door closure (Lemma 2) is the next Lean target. The  $M_{31}$  integers above are thin computational certificates, not Lean theorems.

## References

- [1] J. Pickett, *The hire graph of the 3-free door*, note, 19th September 2026.